

MAT 3007 – Optimization

Interior Point Method in LP and Intro to Nonlinear Optimization

Lecture 15

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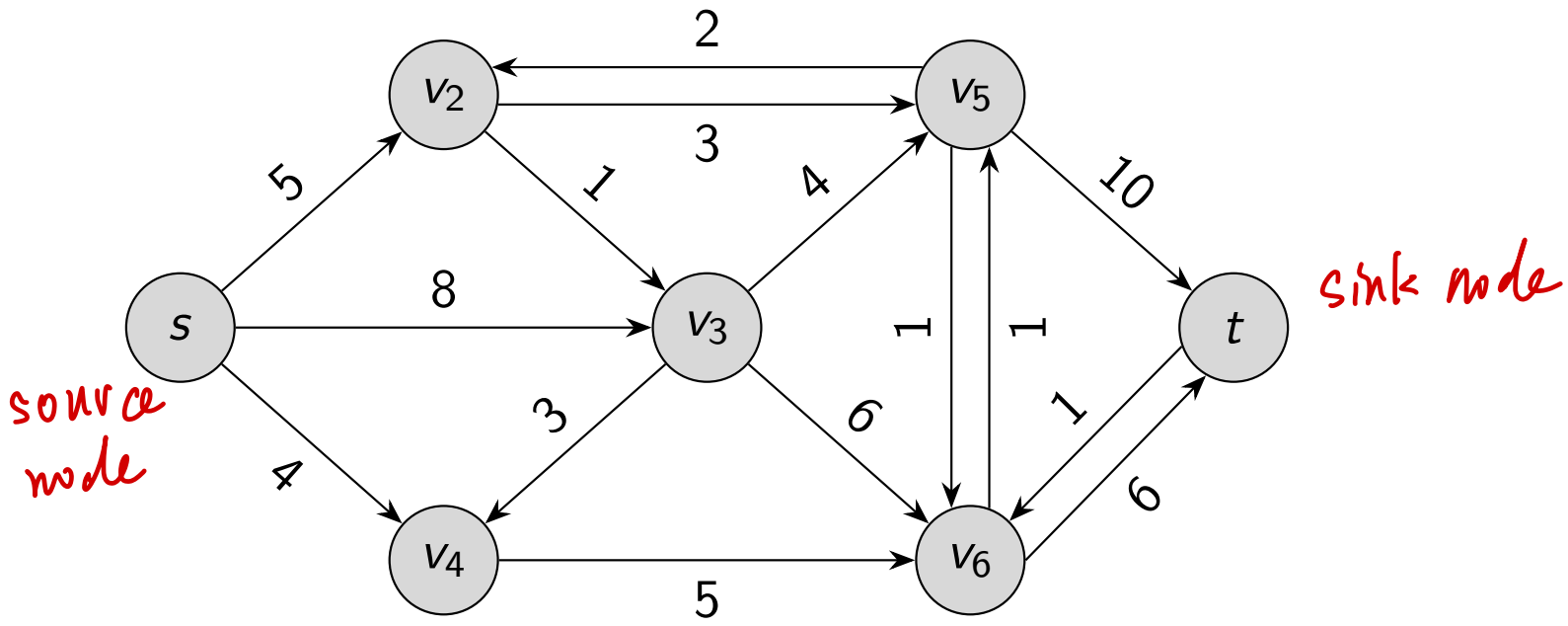
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Recap

Maximum Flow Problem

The maximum flow problem can be described as follows:

- Given a directed, weighted graph $G = (V, E)$ and a pair of nodes s and t (V: set of nodes; E: set of edges).
- There is an edge capacity c_{ij} on each edge.
- **Question:** What is the largest amount of flow one can send from s to t subject to the capacity constraints?



LP Formulation

Using this transformation, we can write down the LP formulation. Let x_{ij} denote the amount of flow across edge (i,j), and Δ be amount of flow from s that ends at t.

dual variable

$$
\begin{aligned}
\max_{x,\Delta} \quad & \Delta \\
\text{s.t.} \quad & \textstyle\sum_{j:(j,i)\in E} x_{ji} - \sum_{j:(i,j)\in E} x_{ij} = 0, && \forall\, i \neq s,t \quad y_i \\
& \textstyle\sum_{j:(j,s)\in E} x_{js} - \sum_{j:(s,j)\in E} x_{sj} + \Delta = 0 && \quad y_s \\
& \textstyle\sum_{j:(j,t)\in E} x_{jt} - \sum_{j:(t,j)\in E} x_{tj} - \Delta = 0 && \quad y_t \\
& x_{ij} \le c_{ij}, && \forall\, (i,j) \in E \quad z_{ij} \\
& x_{ij} \ge 0, && \forall\, (i,j) \in E.
\end{aligned}
$$

flow goes out
= flow goes in + Δ

flow goes out
= flow goes in − Δ

- The first constraint is the flow balancing constraints for all nodes other than s and t.
- The second (third, resp.) constraint is the flow balancing constraints for node s (t, resp.).

Dual of the Maximum Flow Problem

We construct the dual problem:

$$\begin{aligned} \text{minimize} \quad & \sum_{(i,j)\in E} c_{ij} z_{ij} \\ \text{subject to} \quad & z_{ij} \geq y_i - y_j, \qquad \forall\ (i,j) \in E \\ & y_s - y_t = 1 \\ & z_{ij} \geq 0 \end{aligned}$$

$\sum_{i\in S,\, j\notin S} c_{ij}$

if $i\in S$, $j\notin S$ then $z_{ij} \geq 1$

o.w. $z_{ij} \geq 0$

$y_s = 1 \Rightarrow s \in S$

$y_t = 0 \Rightarrow t \notin S$

What does the dual problem mean?

First assume all y_i's are either 0 or 1. Then:

Then let's say node $i \in S$ if $y_i = 1$, $i \notin S$ if $y_i = 0$

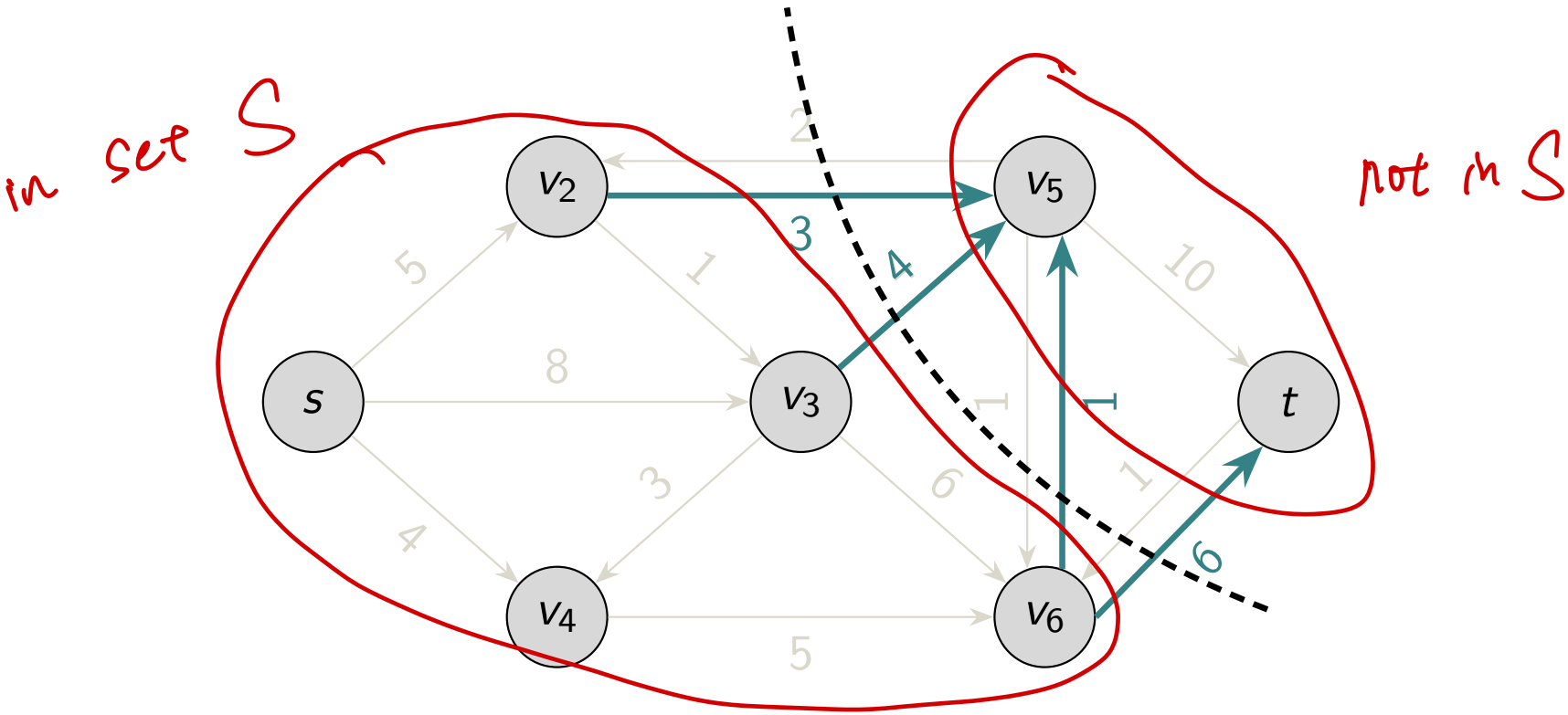
- We assign a label (0 or 1) to each node; 1 to s and 0 to t.
- If i has a larger label than j for $(i,j) \in E$, there is a cost c_{ij}.

Interpretation of the Dual

The dual problem corresponds to finding a subset S of vertices that includes s but excludes t, such that the total weight of the cut, defined as

$$\sum_{i \in S, j \notin S} c_{ij}$$

is minimized. This is known as min-cut problem.



Strong Duality Says ...

Theorem: Max-Flow & Min-Cut

The maximum flow of the network is equal to the smallest cut size of any subset S of vertices.

Corollary

If the value of a flow is equal to the value of some cut, then both are optimal.

- One can view the min-cut as the bottleneck of the network.
- The maximum flow that can be sent through this network is equal to the tightest bottleneck of this network.

This is one classical example of dual problems: maximum flow versus minimum cut.

Global Sensitivity

We now study what will happen if:

1. $\boldsymbol{b}$ changes to $\boldsymbol{b} + \Delta\boldsymbol{b}$
2. $\boldsymbol{c}$ changes to $\boldsymbol{c} + \Delta\boldsymbol{c}$

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1. $\boldsymbol{b}$ changes to $\boldsymbol{b} + \Delta \boldsymbol{b}$
2. $\boldsymbol{c}$ changes to $\boldsymbol{c} + \Delta \boldsymbol{c}$

Recall the simplex tableau:

$\boldsymbol{c}^\top - \boldsymbol{c}_B^\top \boldsymbol{A}_B^{-1} \boldsymbol{A}$	$-\boldsymbol{c}_B^\top \boldsymbol{A}_B^{-1} \boldsymbol{b}$
$\boldsymbol{A}_B^{-1} \boldsymbol{A}$	$\boldsymbol{A}_B^{-1} \boldsymbol{b}$

At the Optimum:

- The reduced costs satisfy $\boldsymbol{c}^\top - \boldsymbol{c}_B^\top \boldsymbol{A}_B^{-1} \boldsymbol{A} \geq 0$.
- $\boldsymbol{A}_B^{-1} \boldsymbol{b}$ and $(\boldsymbol{A}_B^{-1})^\top \boldsymbol{c}_B$ are the basic part of the optimal primal solution and the optimal dual solution, respectively.

Changing $\boldsymbol{b}$

Changing $\boldsymbol{b}$

new problem $\min c^T x$ s.t. $Ax = \tilde{b}$

Suppose $\boldsymbol{b}$ becomes $\tilde{\boldsymbol{b}} = \boldsymbol{b} + \Delta\boldsymbol{b}$. Now, the new basic solution corresponding to the original optimal basis is: $x \geq 0$ to the new problem

$$\tilde{\boldsymbol{x}}_B = \boldsymbol{A}_B^{-1}(\boldsymbol{b} + \Delta\boldsymbol{b}) = \boldsymbol{x}_B^* + \boldsymbol{A}_B^{-1}\Delta\boldsymbol{b}.$$

$\boldsymbol{A}_B^{-1}$: optimal basis for old problem

new basic solution $\tilde{x} = (\tilde{x}_B ; 0)$.

Changing b

Suppose $\boldsymbol{b}$ becomes $\tilde{\boldsymbol{b}} = \boldsymbol{b} + \Delta \boldsymbol{b}$. Now, the new basic solution corresponding to the original optimal basis is:

$$\tilde{\boldsymbol{x}}_B = \boldsymbol{A}_B^{-1}(\boldsymbol{b} + \Delta \boldsymbol{b}) = \boldsymbol{x}_B^* + \boldsymbol{A}_B^{-1}\Delta \boldsymbol{b}.$$

Note that the reduced costs $\boldsymbol{c}^\top - \boldsymbol{c}_B^\top \boldsymbol{A}_B^{-1}\boldsymbol{A}$ do not depend on $\boldsymbol{b}$. Therefore, if $\tilde{\boldsymbol{x}} = (\tilde{\boldsymbol{x}}_B; \boldsymbol{0})$ is feasible, then it is still optimal.

What ranges of changes Δb qualify for a small change such that the local sensitivity analysis hold?

- $\boldsymbol{A}\tilde{\boldsymbol{x}} = \tilde{\boldsymbol{b}}$ still holds since $\tilde{\boldsymbol{x}}$ is a basic solution.
- Need to check $\tilde{\boldsymbol{x}} \geq 0$, i.e., $\tilde{\boldsymbol{x}}_B = \boldsymbol{x}_B^* + \boldsymbol{A}_B^{-1}\Delta \boldsymbol{b} \geq 0$

only thing we need to check.

old optimal basic variable

Changing $\boldsymbol{b}$

Consider the change only occurs in one component of $\boldsymbol{b}$: $\Delta \boldsymbol{b} = \lambda \boldsymbol{e}_i$ ($\boldsymbol{e}_i$ is a vector with 1 at position i). Then, we need to have:

$$\boldsymbol{x}_B^* + \lambda \boldsymbol{A}_B^{-1} \boldsymbol{e}_i \geq 0$$

so that the optimal basis remains the same.

$\rightsquigarrow$ We can find the range of λ by solving these inequalities.

not always easy to find from tableau. If not, we can do inversion by hand.

Example: Production Planning

Consider the production example:

$$
\begin{array}{lrrl}
\text{maximize} & x_1 & +2x_2 & \\
\text{subject to} & x_1 & & \leq 100 \\
& & 2x_2 & \leq 200 \\
& x_1 & +x_2 & \leq 150 \\
& x_1, & x_2 & \geq 0
\end{array}
$$

The optimal basis is $\{1, 2, 3\}$ and we have $\boldsymbol{x}^* = (50, 100, 50, 0, 0)^\top$.

- How much can we change the third right-hand-side coefficient (150) such that the optimal basis remains the same?

Example: Continued

The final simplex tableau is

	x_1	x_2	s_1	s_2	s_3	
B	0	0	0	1/2	1	250
1	1	0	0	-1/2	1	50
3	0	0	1	1/2	-1	50
2	0	1	0	1/2	0	100

Thus $\boldsymbol{A}_B^{-1} = \begin{bmatrix} 0 & -0.5 & 1 \\ 1 & 0.5 & -1 \\ 0 & 0.5 & 0 \end{bmatrix}$.

this is $A_B^{-1} I = A_B^{-1}$ because the part A corresponding to slack variables is identity matrix I_m.

when the standard form LP is derived by adding slack variable, we can read A_B^{-1} from tableau.

Example: Continued

The final simplex tableau is

B	0	0	0	1/2	1	250
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2	0	1	0	1/2	0	100

Thus $\boldsymbol{A}_B^{-1} = \begin{bmatrix} 0 & -0.5 & 1 \\ 1 & 0.5 & -1 \\ 0 & 0.5 & 0 \end{bmatrix}$. If $\boldsymbol{b}$ changes to $\boldsymbol{b} + \lambda \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$,

$$\tilde{\boldsymbol{x}}_B = \boldsymbol{x}_B^* + \lambda \boldsymbol{A}_B^{-1} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 50 \\ 50 \\ 100 \end{bmatrix} + \lambda \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} \geq 0.$$

In order for this to be positive, we need $-50 \leq \lambda \leq 50$.

Another Case that we may find A_B^{-1} without doing matrix inversion explicitly.

Suppose we already have an initial Basis B_0 and $A_{B_0}^{-1}$.

Then the initial tableau corresponding to this B_0 should contain

$$A_{B_0}^{-1} A = A_{B_0}^{-1} [A_{B_0} \quad A_{N_0}] = [I \quad A_{B_0}^{-1} A_{N_0}].$$

If we reach the final optimal solution with basis B, the tableau should contain

$$A_B^{-1} A = A_B^{-1} A_{B_0} A_{B_0}^{-1} A = A_B^{-1} A_{B_0} [I \quad A_{B_0}^{-1} A_{N_0}]$$

$$= [A_B^{-1} A_{B_0} \quad A_B^{-1} A_{N_0}]$$

M ← we can observe this matrix from tableau

this is the part corresponds to initial B_0

$$\Rightarrow M = A_B^{-1} A_{B_0} \iff A_B^{-1} = M A_{B_0}^{-1}$$

i.e we can compute A_B^{-1} as the product of two matrices without explicitly perform matrix inversion

When B_0 and $A_{B_0}^{-1}$ is available ?

If we use two-phase method and B_0 is the initial basis for phase II, B_0 is the basis in the final tableau of phase I. This final tableau also contains $A_{B_0}^{-1}$, because in phase I the part of A corresponding to the auxiliary variable is identity matrix and therefore the final tablea of phase I includes $A_{B_0}^{-1} I_m = A_{B_0}^{-1}$.

Changing $\boldsymbol{c}$

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Now suppose $\boldsymbol{c}$ changes to $\tilde{\boldsymbol{c}} = \boldsymbol{c} + \Delta\boldsymbol{c}$. Because feasibility does not depend on $\boldsymbol{c}$, the orginal solution $\boldsymbol{x}^*$ is still a BFS in the new problem.

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- In order for this BFS to be still optimal, we need to guarantee that the reduced costs are nonnegative.
- We only need to consider the non-basic part since the basic part must still be $\mathbf{0}$:

$$\tilde{\boldsymbol{c}}_N^\top - \tilde{\boldsymbol{c}}_B^\top \boldsymbol{A}_B^{-1}\boldsymbol{A}_N \geq \mathbf{0} \iff (\boldsymbol{c}_N + \Delta\boldsymbol{c}_N)^\top - (\boldsymbol{c}_B + \Delta\boldsymbol{c}_B)^\top \boldsymbol{A}_B^{-1}\boldsymbol{A}_N \geq \mathbf{0}$$

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Assume $\Delta\boldsymbol{c} = \lambda\boldsymbol{e}_j$. We discuss two cases: $j \in B$ and $j \in N$. We study how to find ranges for λ such that the original basis is still optimal (so we can apply the local sensitivity analysis).

Case 1: $j \in B$

we should check what position j is in the basis.

Assume $j = B(\ell)$. In this case, the reduced costs are:

$$\begin{aligned} & \boldsymbol{c}_N^\top - (\boldsymbol{c}_B^\top + \lambda \boldsymbol{e}_\ell^\top) \boldsymbol{A}_B^{-1} \boldsymbol{A}_N \\ = \; & \boldsymbol{c}_N^\top - \boldsymbol{c}_B^\top \boldsymbol{A}_B^{-1} \boldsymbol{A}_N - \lambda \boldsymbol{e}_\ell^\top \boldsymbol{A}_B^{-1} \boldsymbol{A}_N. \end{aligned}$$

Note that $\boldsymbol{c}_N^\top - \boldsymbol{c}_B^\top \boldsymbol{A}_B^{-1} \boldsymbol{A}_N$ are the reduced costs for the original problem. We denote it by $\bar{\boldsymbol{c}}_N^\top$.

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Note that $\boldsymbol{c}_N^\top - \boldsymbol{c}_B^\top \boldsymbol{A}_B^{-1} \boldsymbol{A}_N$ are the reduced costs for the original problem. We denote it by $\bar{\boldsymbol{c}}_N^\top$.

Therefore, in order to maintain the optimality of the current basis, we need to have:

$$\bar{\boldsymbol{c}}_N^\top - \lambda \boldsymbol{e}_\ell^\top \boldsymbol{A}_B^{-1} \boldsymbol{A}_N \geq \boldsymbol{0}. \tag{1}$$

Case 2: $j \in N$

Assume $j = N(\ell)$. In this case, the reduced costs are:

$$\boldsymbol{c}_N^\top + \lambda \boldsymbol{e}_\ell^\top - \boldsymbol{c}_B^\top \boldsymbol{A}_B^{-1} \boldsymbol{A}_N = \bar{\boldsymbol{c}}_N^\top + \lambda \boldsymbol{e}_\ell^\top.$$

Therefore, in order to maintain the optimality of the current basis, we need to have:

$$\bar{\boldsymbol{c}}_N + \lambda \boldsymbol{e}_\ell \geq \boldsymbol{0}. \tag{2}$$

- We can solve the range of λ from (2).

Example: Production Planning

Consider the same production example:

$$
\begin{array}{rrrl}
\text{maximize} & x_1 & +2x_2 & \\
\text{subject to} & x_1 & & \leq 100 \\
& & 2x_2 & \leq 200 \\
& x_1 & +x_2 & \leq 150 \\
& x_1, & x_2 & \geq 0
\end{array}
$$

The final simplex tableau is:

B	0	0	0	1/2	1	250
1	1	0	0	-1/2	1	50
3	0	0	1	1/2	-1	50
2	0	1	0	1/2	0	100

How much can we change the first objective coefficient so that we can use the local sensitivity analysis?

Example: Continued

We have $\bar{\boldsymbol{c}}_N = (0.5, 1)^\top$;

both of them can be read from tableau

$$\boldsymbol{A}_B^{-1}\boldsymbol{A}_N = \begin{bmatrix} -0.5 & 1 \\ 0.5 & -1 \\ 0.5 & 0 \end{bmatrix}.$$

Assume we change the profit 1 from 1 to $1+\lambda$ (i.e., $-1-\lambda$ in the standard form). Then, we need:

$$\bar{\boldsymbol{c}}_N - \lambda(\boldsymbol{A}_B^{-1}\boldsymbol{A}_N)^\top \begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0.5 \\ 1 \end{bmatrix} - \lambda \begin{bmatrix} 0.5 \\ -1 \end{bmatrix} \geq 0$$

$\rightsquigarrow -1 \leq \lambda \leq 1.$

how to determine e_ℓ?

here we change c_j with $j=1$. From the tableau, $B=\{1,3,2\}$, so $B(1)=1$ i.e. $\ell=1$.

Example: Continued

We have $\bar{\boldsymbol{c}}_N = (0.5, 1)^\top$;

$$\boldsymbol{A}_B^{-1}\boldsymbol{A}_N = \begin{bmatrix} -0.5 & 1 \\ 0.5 & -1 \\ 0.5 & 0 \end{bmatrix}.$$

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$\rightsquigarrow -1 \leq \lambda \leq 1.$

- If the profit coefficient of the first product is between 0 and 2, we can use the local sensitivity theorem to compute the opt. value using $\boldsymbol{x}^*$.

Changing $\boldsymbol{A}$

If the change appears in a non-basic column, say in $\boldsymbol{A}_j$, then the original optimal solution is still feasible.

The only change occurs in the reduced costs of jth variable.

- Recompute $\bar{c}_j$. If it is still nonnegative, then the original optimal solution stays optimal. Otherwise, update the tableau for the jth column as well as the reduced cost and continue from there.

If the change appears in a basic column, then nearly all numbers in the tableau will change. In general, there is not a simple way to deal with it.

Other Changes

Adding a Variable (the rest are kept the same):

$$\min \sum_{i=1}^{n} c_i x_i + c_{n+1} x_{n+1}$$
$$\text{s.t.} \quad Ax + \begin{pmatrix} \vdots \\ a \end{pmatrix} x_{n+1} = b$$
$$x \geq 0, \; x_{n+1} \geq 0$$

- The original BFS is still a BFS, the reduced costs are unchanged.
- We only need to check the reduced cost corresponding to the new variable.
- If it is nonnegative, then the original optimal solution is still optimal; otherwise continue the simplex method from there.

Adding a Constraint:

- If the original optimal solution satisfies the constraint, then it is still optimal.
- If not, then the best way to deal with it is to interpret it as adding a dual variable. Then use the simplex tableau for the dual problem to continue calculations.

Example: Adding a Variable

$$
\begin{array}{rrrrl}
\text{maximize} & x_1 & +2x_2 & & \\
\text{subject to} & x_1 & & & \leq 100 \\
& & & 2x_2 & \leq 200 \\
& x_1 & & +x_2 & \leq 150 \\
& x_1, & & x_2 & \geq 0
\end{array}
$$

The optimal solution is $\boldsymbol{x}^* = (50, 100)$ with optimal value 250.

Suppose the company introduces a third product x_3 which requires 1 unit of each resource and can be sold for 2\$:

$$
\begin{array}{rrrrrl}
\text{maximize} & x_1 & +2x_2 & +2x_3 & & \\
\text{subject to} & x_1 & & & +x_3 & \leq 100 \\
& & & 2x_2 & +x_3 & \leq 200 \\
& x_1 & & +x_2 & +x_3 & \leq 150 \\
& x_1, & & x_2, & x_3 & \geq 0
\end{array}
$$

Will this change the optimal solution and value?

Example: Continued

The final simplex tableau is

	x_1	x_2	s_1	s_2	s_3	
B	0	0	0	1/2	1	250
1	1	0	0	-1/2	1	50
3	0	0	1	1/2	-1	50
2	0	1	0	1/2	0	100

- We treat x_3 is a new (6th) non-basic variable $\rightsquigarrow x_3 = 0$.

Current optimal solution $\begin{pmatrix} 50 \\ 100 \\ 50 \\ 0 \\ 0 \end{pmatrix}$

Now consider a new BFS to the new problem. $\begin{pmatrix} 50 \\ 100 \\ 50 \\ 0 \\ 0 \\ 0 \end{pmatrix}$ ← x_3

Example: Continued

The final simplex tableau is

B	0	0	0	1/2	1	250
1	1	0	0	-1/2	1	50
3	0	0	1	1/2	-1	50
2	0	1	0	1/2	0	100

- We treat x_3 is a new (6th) non-basic variable $\rightsquigarrow x_3 = 0$.
- The reduced costs for x_3 are:

$$\bar{c}_6 = c_6 - \boldsymbol{c}_B^\top \boldsymbol{A}_B^{-1} \boldsymbol{A}_6$$

$$= -2 - \begin{bmatrix} -1 & 0 & -2 \end{bmatrix} \begin{bmatrix} 0 & -0.5 & 1 \\ 1 & 0.5 & -1 \\ 0 & 0.5 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = -\frac{1}{2}.$$

We need to update the tableau and continue the simplex method.

Example: Continued

The new simplex tableau is

B	0	0	0	1/2	1	-1/2	250
1	1	0	0	-1/2	1	1/2	50
3	0	0	1	1/2	-1	1/2	50
2	0	1	0	1/2	0	1/2	100

A_B^{-1}

$A_B^{-1}A_b$. Here we can read A_B^{-1} from tableau because the standard form comes from adding slack variables.

Example: Continued

The new simplex tableau is

B	0	0	0	1/2	1	-1/2	250
1	1	0	0	-1/2	1	1/2	50
3	0	0	1	1/2	-1	1/2	50
2	0	1	0	1/2	0	1/2	100

After one step, the final simplex tableau is given by

B	1	0	0	0	2	0	300
6	2	0	0	-1	2	1	100
3	-1	0	1	1	-2	0	0
2	-1	1	0	1	-1	0	50

The new optimal solution is $\boldsymbol{x}^* = (0, 50, 100)$ with profit 300.

Interior Point Method

Start: A New Perspective on Solving LPs

- We now consider the simplex method from a new perspective.

An optimality conditions for LPs (in standard form):

1. Primal Feasibility: $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$, $\boldsymbol{x} \geq \boldsymbol{0}$.
2. Dual Feasibility: $\boldsymbol{A}^\top \boldsymbol{y} \leq \boldsymbol{c}$.
3. Complementarity: $x_i \cdot s_i = x_i \cdot (c_i - \boldsymbol{A}_i^\top \boldsymbol{y}) = 0$ for each i.

These conditions are necessary and sufficient for $\boldsymbol{x}$ and $\boldsymbol{y}$ being the optimal solutions of the primal and dual problems.

Review of the Simplex Method

In the simplex method, we search among basic feasible solutions:

- $\rightsquigarrow$ We always maintain primal feasibility.

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$\rightsquigarrow$ We always maintain primal feasibility.

For any basis B, consider $\boldsymbol{y} = (\boldsymbol{A}_B^{-1})^\top \boldsymbol{c}_B$ as the current dual variable. Note that the reduced costs are $\boldsymbol{c}^\top - \boldsymbol{y}^\top \boldsymbol{A}$. For basic variables, the reduced costs are zero.

Consequently, the complementarity condition is always satisfied:

$$x_i \cdot (c_i - \boldsymbol{A}_i^\top \boldsymbol{y}) = 0 \quad \forall\ i.$$

$=0$ for nonbasic $\quad$ $=0$ for basic variable.

Review of the Simplex Method

During the simplex method, the reduced costs may be negative:

- Hence, the dual solution $\boldsymbol{y} = (\boldsymbol{A}_B^{-1})^\top \boldsymbol{c}_B$ may not always satisfy the constraint $\boldsymbol{A}^\top \boldsymbol{y} \leq \boldsymbol{c}$.
- The method stops whenever the reduced costs are nonnegative.
- We seek $\boldsymbol{y}$ satisfying $\boldsymbol{A}^\top \boldsymbol{y} \leq \boldsymbol{c}$, i.e., dual feasibility.

Proposition: Optimality Properties of the Simplex Method

During each iteration, the simplex method maintains primal feasibility and the complementarity conditions. It seeks a solution that is dual feasible.

What if we choose to maintain the other two conditions?

- This will result in the dual-simplex method and the interior point method. a version called primal-dual interior-point method.

The Dual Simplex Method

One can view the dual simplex method as a simplex method applied to the dual problem of an LP.

- It maintains dual feasibility.
- It maintains the complementarity conditions.
- However, primal feasibility does not need to be satisfied during the iterations. It seeks for primal feasible BFS.

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One can view the dual simplex method as a simplex method applied to the dual problem of an LP.

- It maintains dual feasibility.
- It maintains the complementarity conditions.
- However, primal feasibility does not need to be satisfied during the iterations. It seeks for primal feasible BFS.

There are cases where using the dual simplex method can be more convenient:

- If a dual BFS is available (but we don't have a primal BFS). We mentioned a scenario in the sensitivity analysis (when $\boldsymbol{b}$ is changed by a large amount or a constraint is added).

Optimality Conditions for LPs:

1. Primal Feasibility: $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$, $\boldsymbol{x} \geq \boldsymbol{0}$.
2. Dual Feasibility: $\boldsymbol{A}^\top \boldsymbol{y} \leq \boldsymbol{c}$.
3. Complementarity: $x_i \cdot s_i = x_i \cdot (c_i - \boldsymbol{A}_i^\top \boldsymbol{y}) = 0$ for each i.

The interior point method maintains primal feasibility and dual feasibility during its iterations and seeks for a pair of solutions that satisfy the complementarity conditions.

High-Level Idea

We want to find $\boldsymbol{x}$, $\boldsymbol{y}$ and $\boldsymbol{s}$ such that:

$$\begin{aligned} \boldsymbol{A}\boldsymbol{x} &= \boldsymbol{b}, \quad \boldsymbol{x} \geq \boldsymbol{0} \\ \boldsymbol{A}^\top \boldsymbol{y} + \boldsymbol{s} &= \boldsymbol{c}, \quad \boldsymbol{s} \geq \boldsymbol{0} \\ x_i \cdot s_i &= 0, \quad \forall\ i. \end{aligned}$$

This is a set of nonlinear equations. It is not obvious how to find a solution of this system.

High-Level Idea: Continued

We consider a relaxed version of the problem.

$$\begin{aligned} \boldsymbol{A}\boldsymbol{x} &= \boldsymbol{b}, \quad \boldsymbol{x} \geq \boldsymbol{0} \\ \boldsymbol{A}^\top \boldsymbol{y} + \boldsymbol{s} &= \boldsymbol{c}, \quad \boldsymbol{s} \geq \boldsymbol{0} \\ x_i \cdot s_i &\leq \mu, \quad \forall\ i. \end{aligned}$$

We call $\mu > 0$ the complementarity gap.

Idea: If we have found a solution for a certain μ, then we want to find a solution for a smaller μ. We keep decreasing μ until we can find a solution of the LP.

- The essential step in the interior point method is to show that this is indeed doable – the approach is similar to Newton's method, which we discuss in the second half of the semester.

Why "Interior Point" Method?

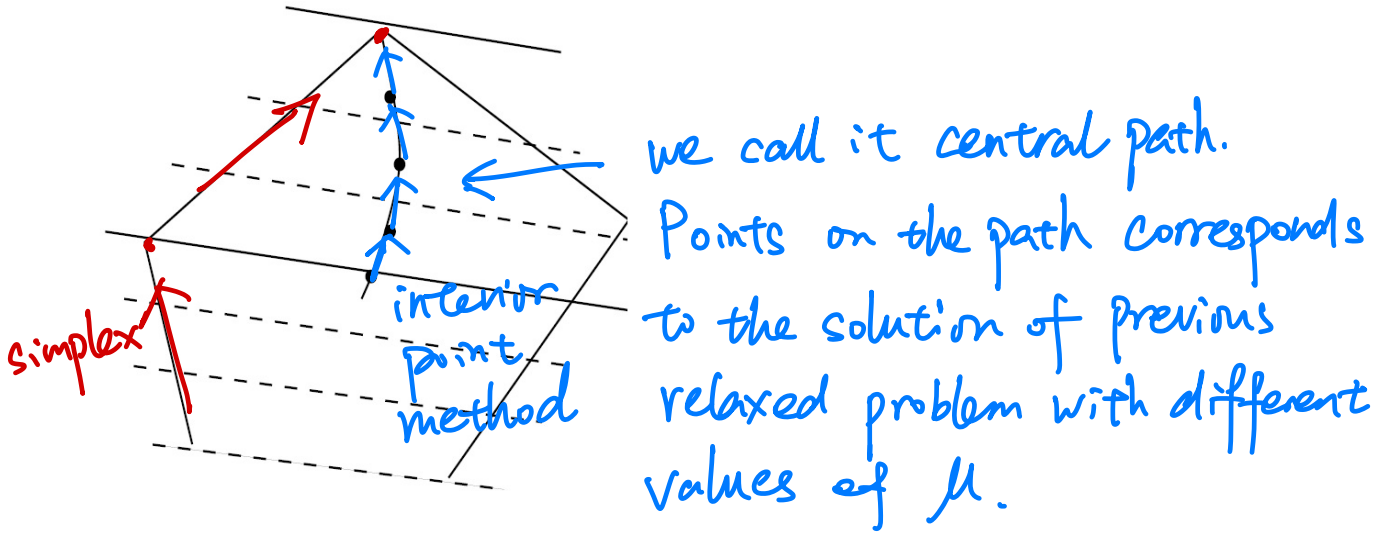


Figure 1: Central path in an interior point method

- In the simplex method, we only search among the extreme points (at the boundary of the polyhedron).
- In the interior point method, we start in the interior of the feasible region.
- Until we reach an optimal solution, we keep $\boldsymbol{x} > 0$ and $\boldsymbol{s} > 0$.
- The optimal solution recovered by the interior point method may not be a BFS (if the solution is unique, it must be a BFS).

A bit more on primal-dual interior point method. (another interpretation)

Consider the following problem

$$\min_x \; c^T x - \mu \sum_{i=1}^{n} \log x_i \qquad \text{s.t} \quad Ax = b$$

$\log x_i$: defined only when $x_i > 0$, and $\to -\infty$ as $x_i \to 0$. So it enforce $x_i > 0$.

$\longrightarrow$ getting closer to the original standard form LP as $\mu \to 0$.

The relaxed problem we prented before is equivalent to an optimality conditim for this nonlinear problem.

Interior Point Method: Initialization & Complexity



We need to find an initial point for the interior point method:

- This can be resolved by solving an auxiliary problem (called the homogeneous self-dual problem).

Interior Point Method: Initialization & Complexity



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Complexity: The interior point method is a polynomial-time algorithm with overall complexity $\approx O(n^{3.5})$.

- There are several variants of the interior point method. The one we introduced is called the primal-dual type of interior point method.
- The main ideas of other variants are similar, i.e., going through the interior of the feasible region and seeking complementarity.

Practical Performance

On average, the speed of the simplex method is comparable with the speed of the interior point method despite their difference in theoretical complexity.

- For some problems, simplex method can be very fast (within a few iterations); for other problems, the simplex method requires some extra time (worst-case exponential).
- In contrast, the running time of the interior point method is quite stable, it does not vary much from problem to problem (given a fixed size).

Property of the Interior Point Method

Informal Statement: Quality of Solutions

The interior point method tends to find dense optimal solution.

Consider a simple case:

$$\begin{aligned} \text{minimize}_x \quad & x_1 + x_2 + x_3 \\ \text{s.t.} \quad & x_1 + x_2 + x_3 = 1 \\ & x_1, x_2, x_3 \geq 0. \end{aligned}$$

- The simplex method will give a BFS as optimal solution (one of $(1,0,0)$, $(0,1,0)$ and $(0,0,1)$).
- The interior point method will return $(1/3, 1/3, 1/3)$.

Discussion

In the case of multiple optimal solutions:

- If we want a dense solution (with many non-zeros), then choose the interior point method.
- If we want a sparse solution (with small number of non-zeros), then choose the simplex method.
- The optimal solution output of the simplex method may depend on the initial solution (as well as on the pivoting rules).

High-Rank vs. Low-Rank

Either situation could be desirable in practice.

Dense Solution:

- In the multi-firm alliance problem, a dense (fair) allocation.

Sparse Solution:

- In graph problems, we want to use fewer nodes/edges.
- In other cases, we prefer integer solutions over fractional solutions (given that their objective values are equal). This usually corresponds to sparse solutions.

Software Choices

Both the simplex method and the interior point method are used in major commercial software:

- In `MATLAB`, we can specify which method we want to use in the function called `linprog`.
- Same in `CPLEX`.
- `CVX` uses the interior point method.
- `Excel` uses the simplex method.

Today's lecture

- Interior Point Method
- Intro to Nonlinear Optimization

Next lecture

- More about Optimality Condition

Questions?